

# Sarp Can Karaman

+90 541 619 9295 | [sarpcankaraman@gmail.com](mailto:sarpcankaraman@gmail.com)

[linkedin.com/in/sarpcankaraman](https://linkedin.com/in/sarpcankaraman) | [github.com/sarpowsky](https://github.com/sarpowsky) | [sarpowsky.github.io](https://sarpowsky.github.io)

## SUMMARY

Third-year Computer Engineering student at Haliç University working in cloud infrastructure and deployment automation. Provisions and operates services on Google Cloud, containerizes workloads with Docker and Kubernetes, and automates build, test, and release through GitHub Actions pipelines with keyless authentication. Comfortable on Linux, and familiar with the Python and data workloads that infrastructure like this typically carries. Currently interning in Cloud & DevOps at Soliner, and serves as Organizer of GDG on Campus Haliç.

## EDUCATION

**Haliç University**

2023 – Present

*B.Sc. Computer Engineering*

## EXPERIENCE

**Soliner**

July 2026 – Present

*Cloud & DevOps Intern*

- Built and managed cloud infrastructure on Google Cloud Platform, handling service provisioning, deployment automation, and environment configuration.
- Designed and implemented CI/CD pipelines with automated testing, linting, and deployment stages.
- Containerized and orchestrated services using Docker and Kubernetes.
- Applied Infrastructure as Code (IaC) principles for configuration management and repeatable environment setup.
- Implemented monitoring and logging for deployed services, and applied cloud networking and security fundamentals with attention to cost optimization.
- Worked on cloud migration and service integration for existing workloads.

**BLUESENSE**

September 2025 – October 2025

*AI Engineer Intern*

- Cleaned and prepared tabular datasets, then developed and improved recommendation system models for the Smart-Beauty mobile application.
- Engineered features and evaluated model performance against real production data.

**Social Office**

September 2025 – October 2025

*IT Intern*

- Developed Python programs for applied business tasks.
- Applied Excel to data analysis and business applications.
- Produced technical designs and drafts using AutoCAD.

**TECHBROS**

July 2025 – August 2025

*Data & AI Intern*

- Completed InternCamp2025, a comprehensive month-long training program covering enterprise systems, data analytics, and emerging technologies; delivered hands-on workshops and technical presentations.
- Trained in DevOps methodologies and deployment pipelines, and in ERP/CRM systems architecture and integration.
- Practiced SQL and NoSQL database management, and machine learning integration in enterprise environments.
- Built skills in data warehousing, ETL processes, data visualization, and statistical analysis techniques.

## PROJECTS

---

### **K8s Carbon Advisor** — *Python, FastAPI, Docker, Kubernetes, GKE, GitHub Actions CI/CD*

- Built and deployed a FastAPI microservice on GKE Autopilot for Kubernetes workload rightsizing (FinOps) and CO<sub>2</sub> footprint estimation, packaged as a multi-stage, non-root Docker image behind a Horizontal Pod Autoscaler.
- Engineered a CI/CD pipeline in GitHub Actions covering automated linting, testing, and keyless GCP authentication via Workload Identity Federation.
- Validated resilience through zero-downtime pod self-healing (250/250 requests served under simulated failure) and confirmed HPA autoscaling behaviour under sustained load.

### **F1 Analytics Warehouse** — *PostgreSQL, dbt, Python, Docker, Metabase*

- Built a containerised PostgreSQL and dbt stack over 7 seasons of Formula 1 timing data (220K lap records), with idempotent upsert ingestion from the FastF1 API so reruns stay safe.
- Implemented 34 dbt models across staging, intermediate, and mart layers with 91 automated tests covering referential integrity, uniqueness, and domain rules.
- Reduced the slowest analytical query from 12 s to 380 ms through composite and BRIN indexing, materialised aggregates, and query restructuring, verified with EXPLAIN ANALYZE plan comparison.

### **MSI Katana RGB Controller** — *Python, PyQt6, HID Protocol, Linux*

- Built a Linux GUI application for hardware control by reverse-engineering the USB HID protocol for keyboard communication.

### **Implicit-Feedback Recommender System** — *Python, PyTorch, Implicit, SciPy Sparse*

- Benchmarked popularity, item-KNN, EASE, ALS, BPR-MF, and a two-tower neural model over 2.7M e-commerce interactions across 90k items, under leave-one-last-item-out and global temporal splits.
- Applied inverse propensity re-ranking to trace the accuracy–coverage frontier, raising catalog coverage from 8% to 31% for a 3.4% NDCG@10 reduction.

### **Uplift Modeling for Campaign Targeting** — *Python, LightGBM, CausalML, EconML, Polars*

- Modelled heterogeneous treatment effects across 13M randomised marketing records, benchmarking S-, T-, X-, and R-learners against uplift forests and causal forests under a shared evaluation protocol.
- Translated model output into an expected-profit curve, identifying an optimal targeting depth of 30% that outperformed blanket treatment by 42% incremental profit under stated cost assumptions.

### **A/B Testing & Experimentation Toolkit** — *Python, NumPy, SciPy, Statsmodels, Streamlit*

- Built an installable experimentation library covering power analysis, sequential testing with alpha spending, CUPED variance reduction, and Bayesian readouts.
- Validated every estimator by Monte Carlo simulation, confirming nominal Type I error, requested-versus-achieved power agreement, and correct confidence-interval coverage across a parameter grid.

## VOLUNTARY

---

### Google Developer Groups on Campus Haliç

*Organizer*

July 2026 – Present

- Took full operational responsibility for community direction, event strategy, and cross-functional team coordination.

*Project Team Lead*

September 2025 – July 2026

- Led the technical operations of the club, overseeing all projects developed within the team.
- Organized Midterm Workshops covering Python, Data Structures, and OOP; co-organized Haliç Engineering Summit'25.

### MultiGroup Community

*Pioneer Member, Data Science Team*

February 2026 – Present

- Contribute to projects and courses organized within Developer MultiGroup, a non-profit software community dedicated to fostering growth, collaboration, and innovation among technology enthusiasts.
- Instruct in the Build With AI workshop series — covering Google Cloud Platform and tools such as Gemini Code Assist — organized with Google for Developers.

## SKILLS

---

**Cloud:** Google Cloud Platform (GKE Autopilot, Cloud Run, GCS, Workload Identity Federation), Service Provisioning, Cloud Migration & Service Integration, Cloud Networking & Security Fundamentals, Cost Optimization

**Containers & Orchestration:** Docker (multi-stage, non-root images), Kubernetes, Horizontal Pod Autoscaling, Container Deployment & Self-Healing

**CI/CD & Automation:** GitHub Actions, Automated Testing & Linting Stages, Keyless Authentication, Infrastructure as Code (IaC), Configuration Management, Monitoring & Logging

**Systems & Programming:** Linux, Bash Scripting, Python (OOP, Pandas, NumPy), Git, FastAPI, USB HID Protocol

**Data:** SQL (PostgreSQL, MySQL, Query Optimization, Database Design), NoSQL, dbt, ETL Pipelines, Data Warehousing

**Machine Learning:** Scikit-learn, XGBoost, LightGBM, PyTorch, Feature Engineering, Model Evaluation

**Tools & Environment:** Streamlit, Metabase, Jupyter Notebook, VS Code, PyCharm